

PROFESSIONAL SUMMARY

Data Scientist specializing in time-series forecasting, predictive modeling, and applied machine learning. Experienced in building end-to-end forecasting and anomaly detection systems using statistical methods (ARIMA, Prophet) and ML models (XGBoost, LSTM). Strong background translating complex datasets into scalable AI solutions deployed in real-world environments.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, C++

Machine Learning: XGBoost, Random Forest, Scikit-learn, Clustering, Anomaly Detection

Deep Learning & Time Series: LSTM, ARIMA, Exponential Smoothing, TensorFlow, YOLOv8

Data & Visualization: Pandas, NumPy, Tableau, ArcGIS

Cloud & MLOps: AWS (EC2, S3, RDS), Azure, GCP, MLflow, Git, SQL Databases

PROFESSIONAL EXPERIENCE

Data Science Researcher - SDSU DiMoLab

Sep 2025 - Present

- Built time-series forecasting models to predict outbreak trends, improving early-risk detection and enabling data-driven public health planning.
- Performed **feature engineering** and outbreak risk modeling.
- Contributed to an AI-driven early-warning framework for public health surveillance.

AI Researcher - Metabolism of Cities Living Lab

Feb 2025 - Dec 2025

- Developed AI tools for indigenous farming communities using **YOLO detection + LLM treatment generation**.
- Integrated **Traditional Ecological Knowledge** with machine learning models.
- Co-authored research **accepted at the 4th International One Health Conference (Rome, 2025)**.

Business Owner - Wholesale Clothing

Apr 2009 - Oct 2016

- Increased sales by **50%** through data-driven market analysis.
- Optimized supply chain through forecasting and analytics.
- Managed import/export operations across Asia.

Process Engineer - ProMOS Technology

Jul 2004 - Feb 2009

- Improved DRAM yield via **statistical process control (SPC)** and root cause analysis.
- Reduced manufacturing defects through data modeling and process optimization.

SELECTED PROJECTS

End-to-End Dengue Forecasting System | Python, LSTM, XGBoost, Random Forest, Streamlit

Built and deployed a time series forecasting system for weekly dengue cases using 15+ years of epidemiological, climate, and vector surveillance data.

- Engineered temporal and lag-based features and compared statistical and ML models to improve forecast accuracy.
- Identified key demand drivers (rainfall, humidity, lagged incidence) to support proactive planning decisions.
- Deployed an interactive forecasting dashboard for real-time exploration and monitoring.

AI-Powered Plant Disease Detection | YOLOv8, LangChain, LLM, Streamlit, Azure

Designed and deployed a production-ready AI system combining computer vision, ML inference, and automated recommendations.

- Built scalable pipelines for model inference and structured knowledge retrieval, demonstrating experience relevant to industrial AI deployment.
- Delivered actionable insights to non-technical users through a web-based interface.

Predicting Restaurant Ratings Using Yelp Metadata | MLflow, XGBoost, RF, Pandas

Built a machine learning pipeline to classify restaurant success (≥ 4 ★ vs < 4 ★) using 36,000+ Yelp business records.

- Engineered 61 features and applied **5-fold cross-validation**.
- Tracked experiments using **MLflow** to compare model performance.
- Optimized **XGBoost**, achieving **0.733 accuracy and 0.79 ROC-AUC**.
- Conducted feature importance and clustering to uncover consumer behavior patterns and restaurant archetypes.

Online Ticket Database Management System | MySQL, AWS RDS, SQL, Python

Developed a full-stack ticket reservation system with cloud-hosted relational databases.

- Designed ERD schema and implemented normalized **MySQL** database on **AWS RDS**.
- Built a Python dashboard for ticket analytics and reporting.
- Automated SQL queries for reservations, seat availability, and revenue statistics.

EDUCATION

M.S. in Big Data Analytics, San Diego State University

Aug 2024 - Dec 2025

Google Data Analytics Professional Certificate

Dec 2022

B.S. in Chemical Engineering, National Cheng Kung University (Taiwan)

Sept 1999 - June 2003